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Securing the internet of robotic things: a comprehensive review on machine learning-based intrusion detection

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ABSTRACT

The Internet of Robotic Things (IoRT) emerges from integrating robotics into the Internet of things (IoT). The dynamic nature of the IoRT environment poses unique security challenges. Implementing adequate security measures can be challenging due to the limited resources, real-time operational requirements, and heterogeneous nature of the mobile network nodes. Machine learning-based intrusion detection systems (IDSs) have significantly improved the security of IoT devices, and other Cyber-Physical Systems (CPSs) by providing embedded intelligence to mitigate cyber threats. Although machine learning models have been extensively used in intrusion detection for IoT systems, their specific application in the context of IoRT remains significantly unexplored. As a result, this paper proposes implementing machine learning-based IDSs to enhance the security of IoRT systems. To provide a comprehensive overview of this emerging research domain, this paper presents a comparative review to differentiate between CPSs, IoT, and IoRT to establish a basis for extending existing machine learning-based IDSs and, the need for more advanced solutions. It discusses the challenges in implementing these security solutions and outlines future research directions. Additionally, the paper introduces an initial concept of a privacy-preserving security framework for IoRT systems.

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KEYWORDS

Internet of robotic things (IoRT); IoT; Cyber-physical systems; intrusion detection systems; machine learning-based intrusion detection

1. Introduction

The Internet of Robotic Things (IoRT) represents a new paradigm of intelligent systems and smart automation, resulting from combining the physical capabilities of autonomous robotic systems and the network and data processing capabilities of the Internet of Things (IoT). Integrating autonomous robotic systems into the IoT ecosystem expands IoT's scope through sensing, actuation, autonomous decision-making, artificial intelligence (AI), and swarm technologies [1].

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This integration allows IoRT systems to collect and process data and physically interact with and manipulate their environment. For instance, IoRT-empowered robots in a supply chain environment can autonomously detect and fix problems in the system, in an attempt to limit human involvement. AI optimizes sensor fusion, extracts patterns from data, and enhances cognition and decision-making processes in IoRT systems. IoT helps to extend robotic systems' functionality through communication, data processing, and collaboration when deployed in dynamic environments [2].

IoRT is a subset of Cyber-Physical Systems (CPSs) that integrates computational algorithms, robotic agents, and feedback control, relying on IoT's network infrastructure. Figure 1 displays the major IoRT capabilities.

Advancements in intelligent systems and smart automation through IoRT introduce some security challenges. The IoRT is a dynamic emerging technology characterized by a constantly evolving environment due to the mobility of robots and the interaction between devices, robots, and humans. It is a complex network of mobile components operating on different protocols, and generating different data. Integrated and sophisticated security solutions capable of handling diverse protocols and their dynamic nature are crucial for seamless operation. However, implementing robust security

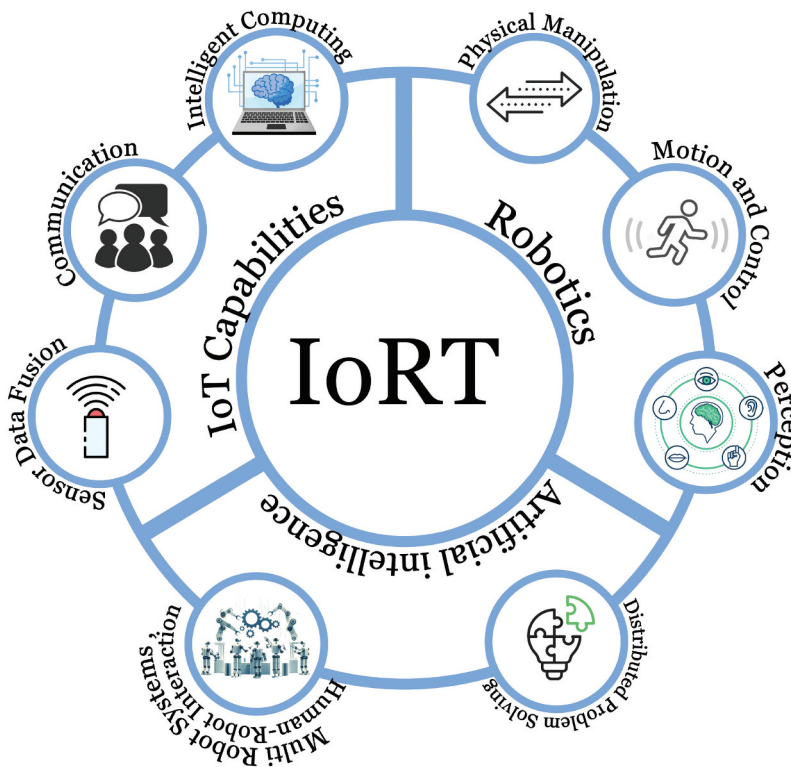


Figure 1. IoRT capabilities.

measures in these networks is challenging due to inherent resource constraints such as limited computational power, communication capabilities, and storage capacities [2].

Conventional security measures such as encryption, authentication, access control, application security, and traditional intrusion detection may be insufficient to secure IoRT systems due to their unique characteristics. Machine learning-based IDSs have successfully helped to secure IoT devices and other CPSs [3]. They offer more effective computing paradigms that address the limitations of traditional intrusion detection approaches and can provide the necessary embedded intelligence for the secure operations of IoRT systems.

While machine learning models have been widely used in intrusion detection for IoT systems, their specific application in the context of IoRT is still an area that requires further research. Our literature review also reveals a gap in research relating to IoRT security issues. Recent works mainly cover a general overview discussing the architecture, components, and challenges of IoRT systems [1,4–6].

This paper employs a systematic methodology to analyze IoRT security vulnerabilities and challenges to provide a comprehensive overview of the emerging research domain. Our methodology is based on reviewing existing research and published literature in the field and synthesis of findings from various studies. It consists of three key phases:

First, we conducted a comprehensive literature review spanning CPSs, IoT, robotics, machine learning-based IDSs, and IoRT published from 2007 to 2024. Our systematic search across Google Scholar, IEEE, Taylor and Francis, Elsevier, and Springer databases used carefully selected keywords and phrases such as Internet of Robotic Things, Cyber-Physical Systems, machine learning-based IDS, and machine learning for security in IoT. From an initial pool of 150 papers, we applied inclusion criteria focused on technical depth, relevance to security challenges, and applicability to IoRT systems, resulting in 106 relevant papers.

Secondly, we compared IoT, CPSs, and IoRT systems across multiple dimensions: physical interaction, decision-making, mobility and navigation, control, and real-time requirements. This comparison helped identify which machine learning-based IDS solutions could be extended to IoRT systems and where new solutions are needed.

Finally, we developed an analytical framework to assess IoRT-specific security challenges, considering factors such as:

- (1) Mobile robot dynamics and their impact on security.
- (2) Resource constraints in IoRT devices.
- (3) Real-time requirements for security solutions.
- (4) Collaborative control implications for security.

Figure 2 displays our research methodology.

Unlike IoT, which primarily focuses on data sharing and communication, IoRT involves real-time decision-making and physical interactions, making its security requirements more complex and critical due to the risk of injury. Based on this analysis, we propose a preliminary privacy-preserving security framework that addresses the unique challenges of IoRT systems while leveraging applicable solutions from IoT and CPS domains. The contributions of this paper are as follows:

- (1) Examines the fundamental concepts of IoRT technology and its vulnerability to cyber threats.
- (2) Presents a taxonomy of IoRT security issues.
- (3) Presents a comparative study that distinguishes between CPSs, IoT, and IoRT.
- (4) Reviews the literature on existing machine learning-based IDSs for IoT, and other CPSs, and explores their extension to IoRT systems.
- (5) Proposes evaluation metrics for machine learning-based IDS customized to IoRT systems.
- (6) Introduces the preliminary concept of a privacy-preserving security framework IoRT systems.

The subsequent sections of the paper are structured as follows: [section 2](#) presents a brief survey of the literature on IoRT. [Section 3](#) presents a comparative study between IoT, CPSs, and IoRT. [Section 4](#) reviews the security challenges in IoRT systems, analyzes their attack surfaces, and discusses potential cyber-physical attacks. [Section 5](#) surveys existing machine learning-based IDSs for IoT and other CPSs. [Section 6](#) discusses the evaluation metrics popularly used in evaluating existing models and proposes some metrics for the IoRT systems. Drawing from our observations, we highlight the challenges encountered and future research directions in [section 7](#). We propose a security framework for IoRT systems in [section 8](#), discuss the shortcomings of our work in [section 9](#), and conclude in [section 10](#).

2. Related works

Recent works on IoRT have provided overviews of its architecture and key components. For example, Afanasyev *et al.* [4] proposed a three-level IoRT reference architecture consisting of a physical layer, a network & control layer, and a service & application layer. Similarly, Ray *et al.* [2] presented IoRT as a five-layer system with layers for hardware or robotics, networking, internet connectivity, infrastructure, and applications. However, these papers do not discuss IoRT security challenges in detail. Afanasyev *et al.* have briefly suggested that

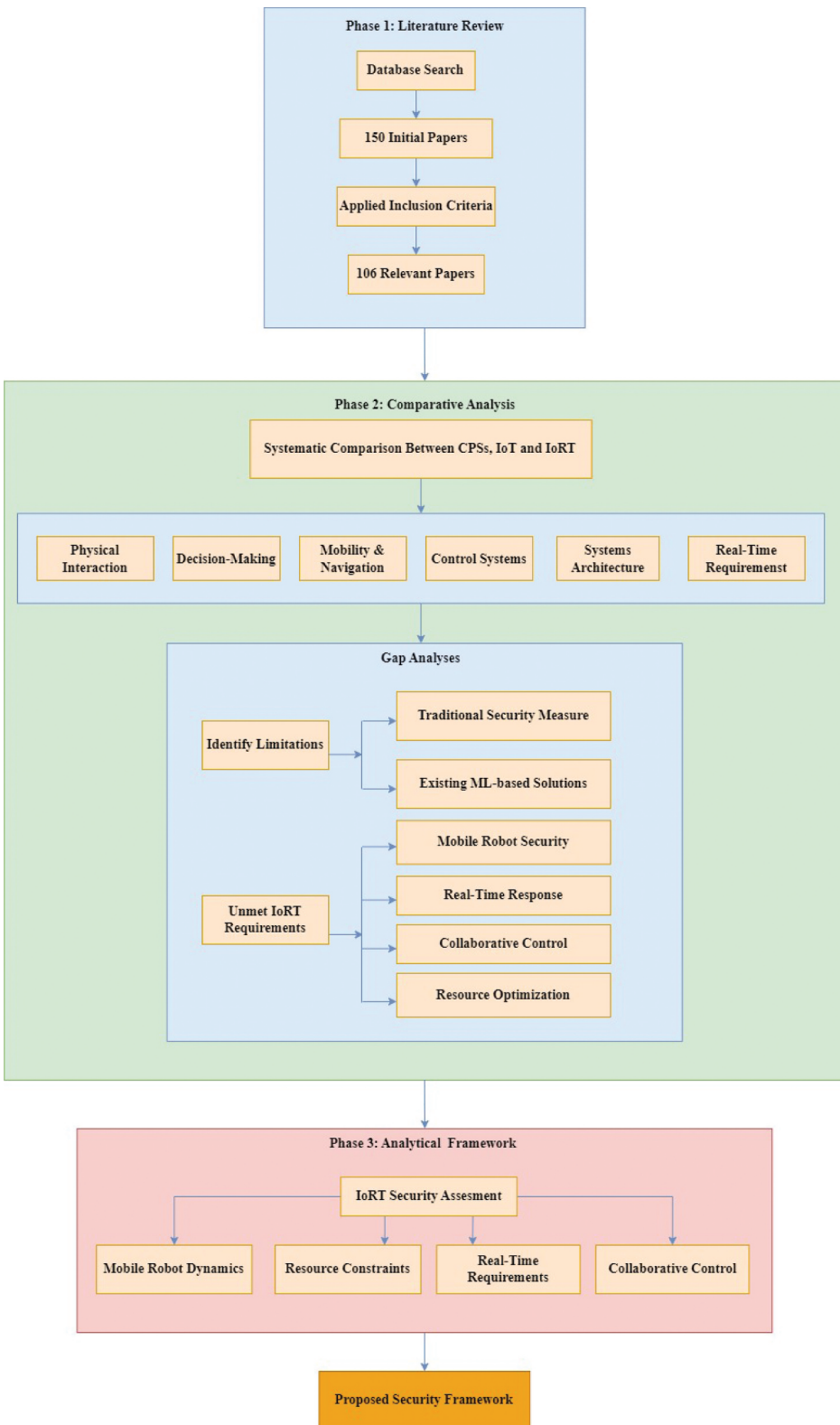


Figure 2. Research methodology leading to IoRT security framework.

IoT security solutions could be applied to IoRT but did not provide a detailed rationale or framework for the extension.

Building on these architectural models, Sayeed and Simeons *et al.* [1,8] have categorized IoRT capabilities into three main levels: basic, higher, and system-level. Basic capabilities include perception, motion, and physical manipulation, while higher-level capabilities involve autonomous decision-making and physical, social, and cognitive interactions. At the system level, key capabilities include configurability, adaptability, and dependability. Sayeed *et al.* [1] further reviewed the technological building blocks that enable IoRT, including robotic technologies like cloud, collaborative, and cognitive robotics, as well as sensors, digital twins, and AI/ML techniques. They have proposed security mechanisms like encryption and trusted data sharing to address the unique challenges in these complex, interconnected systems.

Expanding on the foundational work, Vermasan *et al.* [6] have reviewed IoRT with a focus on intelligent connectivity, architectures, interoperability, and trustworthy frameworks. They examined the technologies that enable IoRT application across different domains. Similarly, Kabir *et al.* [7] provided an overview of the intersection between IoT and Multi-Robot Systems, examining technologies such as navigation, network protocols, and communications.

Looking at the broader context, Romeo *et al.* [11] discussed how key Industry 4.0 technologies have led to IoRT systems development. They outlined various IoRT applications in smart domains and highlighted important issues such as robot-robot and human-robot interactions, energy management, and cybersecurity. Meanwhile, Nayyar *et al.* [9] reviewed IoRT architecture and the technologies required for its development, including existing robot types. They listed existing humanoid, mobile, flying, and swarm robots envisaged for the next-generation IoRT standard. They suggested the need for IoRT security frameworks to transmit information with integrity and confidentiality, though they did not propose a specific framework.

ElBanhawy *et al.* [15] have focused on IoRT use cases in areas like autonomous supply chains, healthcare, personal robots, physical security and monitoring, and automated vehicles. They also offer insights into cloud robotics architecture and computing models, which enable distributed robotic capabilities.

Building on this, the literature has highlighted significant integration of IoT and robotics across multiple domains, particularly in agriculture and healthcare. Eze [17] surveyed current research showing how the convergence between these two domains is advancing IoRT development. Automated robots leveraging embedded coding and IoT connectivity have been deployed in agriculture for plowing, seeding, and irrigation management. Similarly, in healthcare, IoT-enabled robots utilize AI and sensor data for real-time monitoring and decision-making. The research also explores technical

advancements in robot cloud computing, distributed perception among collaborative robots, and communication protocols like ROSLink for seamless IoT-robotic integration.

Looking at security, Zafir *et al.* [16] have emphasized the importance of encryption and blockchain technology in protecting sensitive data transmissions and providing tamper-proof storage for IoRT systems through decentralized ledgers.

The existing literature offers an overview of IoRT applications, enabling technologies, and emerging challenges. A summary of the key related works is provided in Table 1.

Table 1. Related works.

Reference	Year	Discussed Topic	IoRT Architecture	Security Challenges
Ray <i>et al.</i> [2]	2016	Architectural Principles, IoRT Characteristics	Five Layers	Trustworthy IoRT Virtual Machines.
Kabir <i>et al.</i> [7]	2017	Multi-Robot Systems, Navigation, Network Protocols, IoRT Communications	Three Layers	Physical, Network and Control, Service and Application Attacks.
Simeons <i>et al.</i> [8]	2018	Technological Challenges	-	Not Security Focused.
Nayyar <i>et al.</i> [9]	2018	IoRT architecture and Technologies	Five,Nine Layers	Security Framework
Razafimandimby <i>et al.</i> [10]	2018	Collective Coverage and Communication Quality	-	Not Security Focused
Afanasyev <i>et al.</i> [4]	2019	IoRT Key Concepts, Architectural Design, Security Challenges	Three Layers	Lack of Proper Encryption, Authentication Issues, Insecure Communication between robots and users.
Vermasan <i>et al.</i> [6]	2020	Enabling Technologies, IoRT Taxonomy, IoRT Platforms.	Eight Layers	Not Security Focused.
Romeo <i>et al.</i> [11]	2020	Industry 4.0 Key Technologies, IoRT Applications in Smart Domains, Robot-Robot Cooperation, Human-Robot Interfaces, Energy Management, Cybersecurity	Three Layers	Insecure User-Robot Communication, Authentication, Data Privacy, Weak Default Robot Configuration
Leila <i>et al.</i> [12]	2020	IoRT System for Healthcare	-	Not Security Focused
Samara <i>et al.</i> [5]	2021	Technological Application, Architectural Design	Three Layers	Insecure Robot Communication, Authentication Failure, Robot Failure.
Bajeh <i>et al.</i> [13]	2021	IoRT Domains, Methods, Applications	Five Layers	Not Security Focused
Sayeed <i>et al.</i> [1]	2022	IoRT Key Concepts, Abilities, Evolution, Applications, Architectural Design, Robotic Navigation, IoRT security, Technical Challenges	Three, four, five, seven Layers	Security Threats in Data Exchange
vermesan <i>et al.</i> [14]	2022	Emerging IoRT Technologies	Five Layers	End to End Operation and Information Technologies Safety and Security Framework
EIBanhawy <i>et al.</i> [15]	2023	IoRT Applications, Cloud Robotics, IoRT Challenges	Three Layers	Location Privacy
Eze [15]	2023	IoRT in Agriculture and Healthcare	-	Security and Privacy
Zafir <i>et al.</i> [16]	2024	Encryption and Blockchain techniques for IoRT Systems	Three and Five Layers	Data Privacy and System Integrity

3. Internet of robotic things (IoRT), cyber-physical systems (CPSs), and internet of things (IoT)

This section establishes the theoretical foundation for our research by examining three key areas. First, we present an overview of IoRT and its enabling technologies. Second, we compare CPSs, IoT, and IoRT to highlight their distinct characteristics and relationships. Finally, we use this analysis to demonstrate why existing machine learning-based IDS need to be extended and enhanced to address the unique security challenges present in IoRT systems. Through this examination, we establish the theoretical basis for developing more advanced security solutions tailored to IoRT systems.

3.1. IoRT overview

IoRT represents a significant advancement in robotics and automation. As defined by ABI Research, IoRT is a concept where intelligent devices can monitor events, fuse a variety of sensor data from different sources, use local and distributed intelligence to determine the best course of action, then act to control or manipulate objects in the physical world, in some cases while physically moving through the world. In IoRT systems, sensor data is processed to enable control, decision-making, and coordination among robotic agents [18].

The evolution of IoRT can be traced through three paradigms:

- (1) Industrial Robotics: Beginning with preprogrammed industrial robot manipulators, which showed limitations when deployed in dynamic environments.
- (2) Networked Robotics: Integrating robotic technologies with existing network infrastructure enabled interconnected robotic devices to communicate over networks like the Internet or LAN [19]. However, this approach faced challenges including limited memory space, inconsistent quality of service, lack of intelligence, and network latency [2].
- (3) Cloud Robotics: Developed to address the limitations of networked robotics, this phase leveraged cloud computing infrastructure to provide robots with access to distributed computing resources and shared training data. However, cloud robotics continued to face interoperability, network latency, security, and standardization challenges.

IoRT emerged as a solution to these challenges by integrating cloud robotics with IoT infrastructure, creating more intelligent, collaborative, heterogeneous, and cost-effective robotic networks [2]. Modern IoRT systems are distinguished by their advanced capabilities in:

- Perception and processing power

- Cognition and intelligence
- Planning and decision-making
- Control and connectivity
- Human-robot interactions (HRIs) [6]

The applications of IoRT span numerous domains where IoT and robotics have traditionally been deployed, including:

- Healthcare and medical services
- Human assistive operations
- Industrial assembly and manufacturing
- Search and rescue operations
- Military applications
- Smart environments (homes, offices, factories, cities)

Recent research has demonstrated novel applications of IoRT technology. For instance, Bajeh *et al.* [13] proposed implementing IoRT in academic libraries for autonomous catalog and collection stack management. In response to the COVID-19 pandemic, Leila *et al.* [12] developed an IoRT system for collecting and transmitting patient data to cloud platforms for remote medical interpretation, helping to control disease spread.

3.2. IoRT architecture

The IoRT architecture consists of three fundamental layers, as illustrated in Figure 3: the physical layer (also known as hardware or perception layer), the network & control layer, and the service & application layer [4,20,21]. Each layer serves distinct functions while working together to enable the complete IoRT system.

Physical Layer: The foundation of the IoRT architecture comprises three key components [2].

- **Sensors:** Monitor and measure physical phenomena, converting them into electrical signals for transmission to control centers.
- **Actuators:** Receive electrical signals from control devices and convert them into physical motion (e.g. moving, gripping, and rotation).
- **Robots:** Autonomous units that perceive their environment and act based on the received information, guided by in-built algorithms.

Network & Control layer: This intermediate layer facilitates integration and communication between physical layer components through

- Hardware components (Controllers, Routers, Servers)

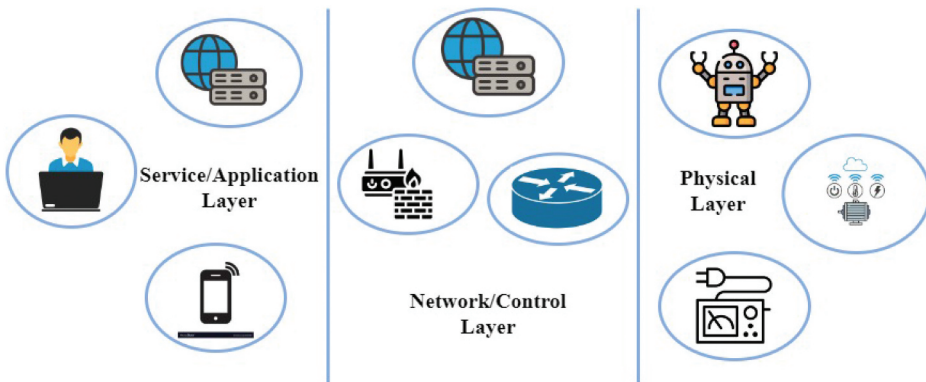


Figure 3. IoRT architecture.

- Communication protocols and technologies (Wi-Fi, 6LoWPAN, Bluetooth Low Energy, Radio Frequency Identification, Near Field Communication, WSN-based communication) [22–25].

These diverse communication methods ensure efficient information exchange across short and long distances within the robotic network infrastructure.

Service & Application Layer: At the highest level, this layer provides

- Software systems for managing, processing, and analyzing environmental variables and agents in smart environments.
- Integration capabilities for AI and machine learning techniques to address complex security challenges [26].
- Application interfaces for end-user interaction and system control.

Through this layered architecture, IoRT systems can effectively combine sensing, communication, and intelligent processing to enable sophisticated robotic applications.

3.3. IoRT enabling technology

IoRT represents a convergence of technologies that enable robots to perform tasks autonomously and collaboratively, interact seamlessly with other devices, and adapt to dynamic environments. Five key enabling technologies form the foundation of IoRT systems: autonomous robotic systems, intelligent computing, AI, IoT, and intelligent connectivity.

3.3.1. Autonomous robotic systems

Autonomous robotic systems are fundamental to IoRT, supporting applications from manufacturing and logistics to healthcare and smart cities. IoRT integrates

technologies that mimic human senses to better interact with humans and other robotic systems. Key capabilities include object transport and manipulation, motion planning, distributed learning, reconfigurable robots, navigation, traffic control, robot formation control, and multi-robot systems. Tactile feedback technologies are also crucial for assistive robotic devices in various IoRT applications.

3.3.2. Intelligent computing

Most autonomous robotic systems traditionally maintain onboard computational and control systems [10]. IoRT extends this by offloading control and computational mechanisms to the cloud leveraging distributed computing power and network connectivity. However, cloud computing alone cannot fully address the latency-sensitivity requirements, computational limitations, and network heterogeneity of IoRT applications. Edge and fog computing resolve these challenges. Edge computing enables data processing directly on devices [27], while fog computing defines how edge computing should be organized and orchestrates computing, storage, and networking resources between cloud and end devices [4]. This approach reduces cloud communication overhead and supports mobile robot operations by bringing computational resources closer to the devices.

3.3.3. Artificial intelligence

AI in IoRT focuses on performing tasks associated with human cognition, enabling robots to plan and execute actions that maximize success [28]. AI encompasses seven primary areas: Knowledge Representation, Natural Language Processing, Learning, Planning and Problem-solving, Inference, Search, and Vision. While AI in traditional IoT primarily analyzes sensor data, IoRT-specific AI enables physical action execution and complex autonomous behavior. This enhanced capability allows individual robotic units to exhibit collaborative fleet-level behaviors [6], providing the foundation for effective operation in dynamic environments.

3.3.4. Intelligent connectivity

Wireless communication technologies form a crucial backbone for IoRT systems, connecting diverse devices and enabling real-time data exchange. Intelligent connectivity combines high-speed, low-latency cellular networks (5G and beyond), AI capabilities, and various IoT devices to create a robust IoRT foundation. Developing resilient wireless or cellular connections is essential for IoRT device and application development. This reliable connectivity enables efficient data exchange, real-time communication, and seamless operation across heterogeneous devices, ultimately enhancing the overall IoRT ecosystem performance [6].

3.4. CPSs and IoRT

CPSs enable digital modeling, automation, and control of physical processes, creating digital representations of systems. These systems employ embedded subsystems and communication networks with feedback loops to monitor and control physical processes in real-time [29]. Given their critical applications, CPSs incorporate redundancy, fault tolerance, and safety mechanisms to ensure reliable operation.

IoRT is a specialized subset of CPSs focused on robotics. It shares fundamental characteristics in integrating computational algorithms, physical devices, and feedback control with CPSs. Both leverage IoT's network infrastructure, but their implementation differs. IoRT encompasses robot manipulator arms and mobile and static robots with advanced sensors, actuators, and connectivity features. CPSs include industrial control and robotic systems, smart grids, and autonomous vehicles with embedded computing capabilities.

CPSs primarily utilize centralized feedback control with dedicated system controllers. IoRT employs complex algorithms for collaborative feedback control, such as behavior-based formation control in robot teams

CPSs often operate in well-defined static environments with centralized resources while IoRT navigates dynamic environments with distributed resources and communication requirements.

The integration of physical and digital components in CPSs and IoRT can be viewed differently: CPSs emphasize the intersection between physical processes and computational components, focusing on their mutual influence rather than their simple combination. This integration fuses various engineering disciplines with computer science models and methods [30]. IoRT can be analyzed from a union and intersection perspective. The union perspective views IoRT as a combination of IoT and robotics, where each maintains distinct characteristics while complementing each other. For example, IoT provides connectivity and sensor data while robotics contributes autonomous capabilities. The intersection perspective emphasizes the seamless integration between robots and IoT, enabling real-time interaction and continuous exchange of data and feedback for dynamic decision-making and action execution.

3.5. IoT and IoRT

The International Telecommunication Union (ITU) defines IoT as a global infrastructure for the information society, enabling advanced services by interconnecting physical and virtual things based on existing and evolving interoperable information and communication technologies. IoT extends beyond Machine-to-Machine communication (M2M) to incorporate ambient intelligence, human-centric interactions, context awareness, and smart environments [31,32].

IoT and IoRT exhibit distinct capabilities and control characteristics. IoT primarily focuses on automation, connectivity, sensing, and data analysis [33]. It Operates as an open-loop control system without real-time feedback requirements. IoT systems execute predefined tasks through embedded systems [34], emphasizing a data-centric approach to monitoring environmental conditions and user behavior. IoRT on the other hand possesses advanced capabilities such as perception, cognition, intelligence, planning, motion, and physical manipulation. They generate trajectories [35], manage energy levels, and perform diagnostics. They employ complex control mechanisms such as feedback, linear, non-linear, feedforward, adaptive, and robust control [36]. Additionally, IoRT requires specialized operating systems (ROS, ROS-2, RTOS) for real-time processing.

While IoT and IoRT share fundamental features in sensor integration, data processing, and communication protocols [8], their security requirements differ significantly. IoT security solutions primarily focus on network-level threat detection through traffic analyses. Its applications have varying timing requirements for security measures meanwhile, IoRT demands consistent real-time security measures due to physical actuation, safety-critical operations, combined mobility, and autonomous decision-making.

Figure 4 illustrates the relationship between CPSs, IoRT, and IoT, with the major differences detailed in Table 2.

4. IoRT security

This section discusses IoRT vulnerabilities, attack surfaces, and some security solutions.

4.1. IoRT vulnerabilities

IoRT systems possess various vulnerabilities that attackers can exploit. Through our literature review, we have developed a two-level classification system for these vulnerabilities.

4.1.1. Infrastructure-level vulnerabilities

Infrastructure-level vulnerabilities concern IoRT deployments. They include:

Physical Components: The physical infrastructure is often deployed in accessible environments where they're susceptible to tampering.

Network Infrastructure: Network Infrastructure vulnerabilities arise from using various wireless communication protocols, each with its security weaknesses.

Data Processing and Storage Infrastructures: Data Processing and Storage Infrastructures present another layer of vulnerability. IoRT systems generate

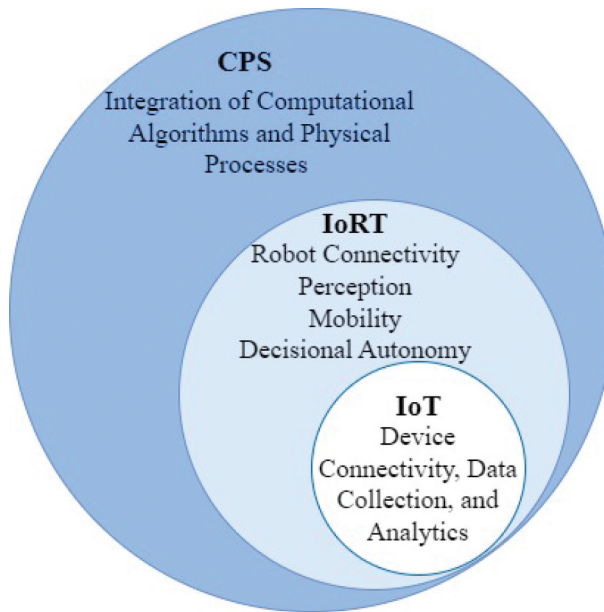


Figure 4. CPS, IoRT, and IoT.

Table 2. Cpss, IoT, and IoRT.

Features	CPSs	IoT	IoRT
Focus	Physical and Computational Process Integration	Device Connectivity, Data Collection and Analysis	Robot Connectivity
Major Components	Physical Processes, Cyber-space, Communication Networks	Sensors, Actuators, Communication Networks, Data Analytic Tools, Smart Devices	Sensors, Actuators, Communication Networks, Robotic Agents
Functionality	Physical Process Control and Monitoring	Task Automation	Autonomy in Dynamic Environments
Control	Centralized Feedback Control	Closed Loop Control	Distributed Feedback Control
Physical Interaction	Process Control and Monitoring	Limited Physical Interactions	Active Manipulation and Autonomous Physical Actions
Decision-Making	Pre-determined Control Loops	Data-Driven Insights	Autonomous Real-Time Decisions
Mobility and Navigation	Fixed or Limited Mobility	Stationary Nodes	Dynamic Mobility and Path Planning
System Architecture	Tightly Coupled Physical-Cyber Components	Distributed Sensor Networks	Hierarchical Robot-Cloud Architecture
Heterogeneity	Physical and Computational Process Diversity, Communication Protocols	Device, Data, IoT Platforms, Communication Protocols	Robot Diversity, IoRT Platforms, Communication Protocols
Communication	Human-to-human, Human-to-Machine, Machine-to-Human, Machine-to-Machine	Device-to-Device, Device-to-Cloud, Device-to-Gateway, Back-End Data Sharing	Human-to-Machine, Human-Robot, Robot-Robot

and process massive amounts of sensitive data, including environmental information, control commands, and personal data. The distributed nature of data storage, often spanning edge devices, fog nodes, and cloud servers, creates

multiple points where data can be intercepted or compromised. Furthermore, many IoRT systems lack robust encryption mechanisms for data in transit and at rest, partly due to resource constraints on edge devices.

Robot Design: Cybersecurity is often not prioritized during robot design, manufacturing, and deployment, leaving room for attackers to attack IoRT systems [37].

IoRT Cloud-Based Platforms: Inherent cloud vulnerabilities such as misconfigurations, unpatched systems, and software bugs may compromise IoRT systems.

Robotic Complexities: Modular and customizable robot designs increase system complexity, expanding attack surfaces and vulnerabilities [37].

Robotic System Setup: Malicious actors can exploit robotic system setup to access networks and systems, enabling theft, modification, or unauthorized access.

Real-time operational requirements: Real-time operational requirements of IoRT systems also introduce vulnerabilities. These systems often need to make split-second decisions based on sensor data and environmental conditions. The need for rapid response times can conflict with security measures that might introduce latency, leading to potential trade-offs between system performance and security.

Authentication and Access Control: The dynamic nature of IoRT systems may require devices to join or leave the network frequently. Traditional authentication mechanisms may not be suitable for resource-constrained devices, leading to potential security compromises.

4.1.2. Architecture-level vulnerabilities

The architectural vulnerabilities in IoRT systems often stem from their inherently distributed and heterogeneous nature. Integrating various protocols, communication standards, and hardware architectures creates compatibility issues and security gaps.

Insecure Communication: Insecure communication between robots, IoT devices, and cloud infrastructure enables attacks such as man-in-the-middle, eavesdropping, sniffing, spoofing, and denial-of-service attacks.

Real Time Location Systems (RTLs): RTLs used for robot navigation can be vulnerable to spoofing and eavesdropping attacks. They often rely on wireless communication technologies for location data transmission.

IoRT Virtual Machine Environment: Virtual machines in IoRT are important for optimized resource allocation, providing isolated environments for enhanced security, and enabling scalable deployments. Lack of robust authentication can allow malicious VMs to impersonate legitimate ones and launch attacks.

Device Heterogeneity: Diverse devices, communication protocols, and security measures across devices create exploitable vulnerabilities.

ROS Vulnerabilities: Lack of encryption and insufficient authentication in ROS can allow attackers to manipulate or intercept data.

Lack of Security Standards: The absence of standardized security frameworks and vulnerability assessments increases IoRT's susceptibility to cyber-attacks.

Figure 5 displays a taxonomy for IoRT security challenges.

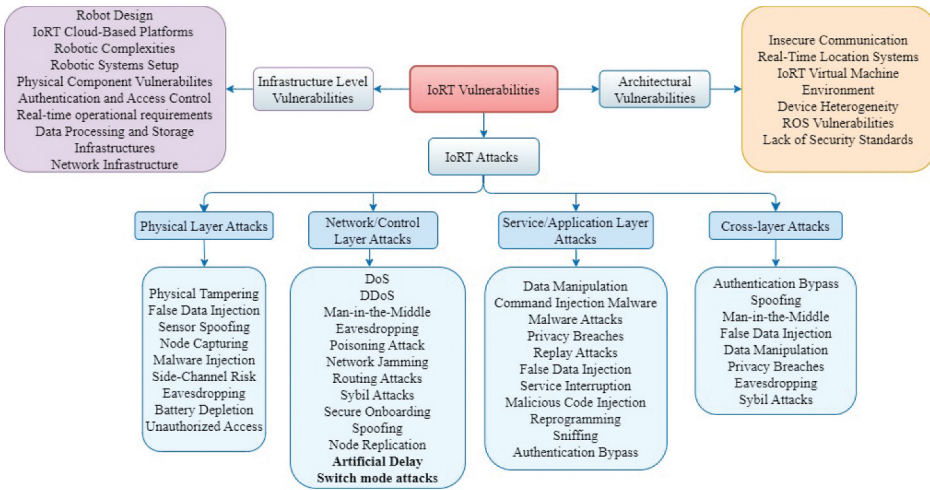


Figure 5. Taxonomy of IoRT security challenges.

4.2. IoRT security threats and attacks

IoRT systems face threats that exploit vulnerabilities across their cyber-physical infrastructure, compromising system assets and jeopardizing the confidentiality, integrity, and availability of systems and data operations. Our analysis synthesized findings from the literature to categorize them into:

4.2.1. Physical layer attacks

IoRT components are vulnerable to physical tampering, especially in unattended environments, allowing intruders to launch attacks. Key threats include false data injection, sensor spoofing, node capturing, malware injection, sink-holes, and Sybil attacks. False data injection targets the sensors to mislead decision-making, while sensor spoofing takes advantage of lax authentication in TCP/IP protocols to impersonate legitimate data sources. Sybil attacks involve creating multiple fake nodes to control a network, and malicious sensor nodes can corrupt sensitive data.

4.2.2. Network & control layer attacks

Communication at the network & control layer is vulnerable to various attacks, including denial-of-service, Sybil, man-in-the-middle, and spoofing. In autonomous robotic vehicles, artificial delay attacks disrupt control systems by delaying data transmission, while switch mode attacks inject false data during mode transitions. Actuators in the IoRT systems face threats like finite energy attacks, finite time attacks with high control signals, and bounded attacks that manipulate actuator behavior.

4.2.3. *Services and applications attacks*

The Service and Application layer in IoRT runs programs that manage and analyze environmental parameters. It's vulnerable to malware (like viruses and trojans), buffer overflow, and several attacks, including Data Stealing, False Data Injection, and malicious code injection. [Figure 5](#) illustrates the taxonomy of IoRT security issues.

IoRT attacks impact data confidentiality, data and resource integrity, and availability of systems and resources. Unauthorized access by malicious actors to sensor nodes in IoRT systems may lead to data breaches, and privacy violations, or affect the reliability and accuracy of data acquisition. Attacks on actuation may lead to system failure or incorrect actions potentially leading to hazardous situations. DoS and DDoS attacks may affect the availability of systems and resources leading to resource exhaustion and delays in robotic systems operations. Unauthorized modifications in robotic firmware may also lead to altered functionality and incorrect behaviors.

Poisoning attacks pose a severe threat by compromising machine learning models and physical robotic actions. Drawing parallels from healthcare machine learning systems [38] where poisoning attacks can manipulate diagnostic outcomes, IoRT faces similar vulnerabilities where compromised sensor data could lead to dangerous physical actions in the real world. Resource constraints in IoRT make them vulnerable to such attacks.

4.3. *IoRT security solutions*

IoRT systems require robust security measures to ensure safe and secure operation. Key approaches include trust-based security using statistical models to identify malicious, and encryption-based mechanisms that ensure data confidentiality, integrity, and availability [1].

While cryptographic methods help secure communication, they are insufficient for the dynamic IoRT. Intrusion detection systems (IDSs) can protect against attacks by analyzing network traffic and system logs to identify attack patterns. There are network, host, signature, and anomaly-based IDSs. Network IDSs detect malicious incoming and outgoing traffic at strategic points within a network, host IDSs are installed on individual devices connected to the internet to detect malicious packets from within. Signature-based IDSs monitor packets on a network and compare them with a database of known attack signatures, anomaly-based IDSs do not rely on predefined signatures, but monitor network traffic and compare them with a predefined normal behavior [39]. Machine learning-based IDSs offer advantages for real-time intrusion detection, reduced false positives, zero-day attack detection, and adaptation to new threats- making them well-suited for dynamic environments [40].

Federated learning-based IDSs further improve detection accuracy while preserving data privacy.

While IoRT shares similarities with IoT, its unique characteristics such as real-time requirements, physical manipulation, power constraints, and mobility may require specialized adaptations of existing ML-based IDSs.

5. Machine learning techniques for IoRT security

Machine learning algorithms analyze data sets to identify meaningful patterns and relationships. These algorithms can be broadly categorized into traditional and advanced deep learning approaches. Traditional machine learning techniques include algorithms such as Random Forests (RF), Decision Trees (DT), Support Vector Machines (SVM), and K-nearest neighbors (KNN), while deep learning methods employ multiple hidden layers in neural networks to extract discriminative or generative feature sets from data [41]. Deep learning's hierarchical architecture distinguishes it from conventional methods, enabling effective feature extraction from large-scale datasets and demonstrating superior learning capabilities. However, these advantages come with trade-offs: deep learning models typically require extensive training periods, and larger datasets to achieve high accuracy and substantial computational resources [42]. Furthermore, they often involve numerous hyperparameters that demand careful tuning to optimize performance and prevent overfitting or underfitting. In contrast, traditional machine learning methods excel at making predictions with smaller datasets and generally require less computational overhead.

Historically, machine learning training in distributed networks has been centralized, with data collected from client nodes processed at a central location. This approach raises significant concerns regarding data privacy and communication efficiency [43]. To address these challenges, Google introduced federated learning [44]. This distributed learning paradigm enables participating nodes to train a global model while maintaining data privacy. Each node trains its local model using local data and shares only model parameters with a central node for aggregation. Recent research has successfully applied federated learning to develop IDSs for IoT and other CPSs, effectively addressing data privacy and protection requirements [45,46].

Machine learning-based IDSs leverage these advanced techniques to develop models capable of detecting cyber-attacks [47]. Compared to traditional rule-based or signature-based approaches, machine learning-based systems offer several advantages, including reduced false positives and the ability to detect zero-day attacks. Their adaptability to evolving attack methods makes them more effective in dynamic environments [40] such as IoRT. However, maintaining optimal performance requires regular assessment and evaluation to identify and address potential limitations.

This section examines existing machine-learning-based IDSs applicable to enhancing IoRT systems security, drawing from successful implementations in IoT devices and CPSs. We categorized these approaches into conventional and deep learning-based IDSs. Each category is subdivided based on learning paradigms: supervised, unsupervised, hybrid, ensemble, semi-supervised, and reinforcement learning methods.

5.1. Supervised learning algorithms

Supervised machine learning algorithms leverage labeled data to train models for prediction or classification tasks. Several supervised learning algorithms have demonstrated effectiveness in IDSs including SVM [48,49], DT [50–52], KNN [53–56], Bayesian algorithms [7,57], and Artificial Neural Networks (ANNs).

Heba and Shams *et al.* [48,49] proposed an SVM-based IDS specifically targeting DoS attacks in MANET networks achieving a 94% detection rate with minimal computational overhead.

Several researchers have explored DT-based approaches. Ingre *et al.* [50] implemented a DT-based IDS using Correlation Feature Selection (CFS) for feature selection for improved accuracy, achieving 90.2% accuracy. Accuracy reduces with an increasing number of classes. Ferrag *et al.* [51] introduced the Rules and DT-based IDS (RDTIDS) for IoT networks, combining DT and rules-based approaches to achieve detection rates 94.475% and 95.175% on the CICIDS_2017 and Bot-IoT datasets, respectively. Moon *et al.* [52] developed a DT-based system for detecting Advanced Persistent Threats (APT) attacks, achieving 84.7% accuracy.

Kabir *et al.* proposed a wrapper-based approach combining feature selection with a Bayesian network, achieving 98.2653% accuracy, and 0.007 false positive rates on the NSL-KDD dataset. Mehmood *et al.* [57] introduced NBC-MAIDS, implementing the Naïve Bayes classification within a multi-agent system for DDoS detection in IoT networks.

The KNN algorithm, a non-parametric technique suitable for numerical and categorical features, has shown promise in various IDS applications. Li *et al.* [53] developed a KNN-based IDS for wireless sensor networks, while Eigner *et al.* [54] applied KNN to detect Man-in-the-Middle attacks in industrial control networks.

ANNs, comprised of interconnected nodes mimicking biological neurons, utilize backpropagation to optimize their predictions [58,59]. They have demonstrated exceptional performance in IoT anomaly detection [65]. Hodo *et al.* [58] achieved 99.4% accuracy in DDoS/DoS attack detection, while Ketzaki *et al.* [60] developed a system capable of detecting diverse threat types.

Convolutional Neural Networks (CNNs), characterized by their convolution, pooling, and fully connected layers, excel at learning hierarchical spatial features [66]. They have been successfully applied to network traffic analysis [63],

malware detection [62], and cybersecurity [61]. Notably, Ren *et al.* [61] combined a CNN with Spectral Residual (SR) for time-series anomaly detection, outperforming existing methods on public and Microsoft datasets. While CNNs require substantial computational resources, their feature extraction capabilities make them valuable for anomaly detection tasks.

Recurrent Neural Networks (RNNs) analyze sequential or time series data through feedback loops that maintain temporal information [67]. Torres *et al.* [64] leveraged RNNs to model IoT network traffic as evolving states, achieving a 99.9% accuracy in attack detection. The sequential nature of RNNs makes them suitable for IoT environments where network traffic patterns evolve. These supervised models are summarized in Table 3.

5.2. Unsupervised learning algorithms

Unsupervised learning algorithms excel at detecting patterns in unlabeled data [68], primarily through clustering and feature extraction techniques. Key unsupervised approaches implemented in IoT systems include Principal Component Analysis (PCA), K-means clustering, autoencoders, Boltzmann Machines, and Deep Belief Networks (DBNs). These methods have demonstrated anomaly, fault, and intrusion detection effectiveness within IoT systems [69].

Table 3. Supervised machine learning-based IDSs.

References	Algorithms	Domain	Datasets	Evaluation Metrics
Shams <i>et al.</i> [49]	SVM	Mobile Ad Hoc Network	Ad hoc Network Data.	F1-score, Mean Packet Delivery Ratio, Average End-to-End Delay.
Ferrag <i>et al.</i> [51]	DT	IoT	CICIDS2017, BoT-IoT	Accuracy, Detection Rate, False Alarm Rate, Time Overhead.
Mehmood <i>et al.</i> [57]	Naïve Bayes	IoT	NSL-KDD	Detection rate, End-to-End Packet-Forwarding Rate, Delay Performance.
Li <i>et al.</i> [53]	KNN	WSN	Network collected data	Detection accuracy, Detection Rate, False Alarm Rate.
Eigner <i>et al.</i> [54]	KNN	Industrial Control Systems	Network Generated Data	False Positives.
Hodo <i>et al.</i> [58]	ANN	IoT	KDD Cup 1999	Accuracy, False Positives, True positives.
Dias <i>et al.</i> [59]	ANN	Computer Networks	KDD Cup 1999	Accuracy.
Ketzaki <i>et al.</i> [60]	ANN	Modern Computer Networks	UNSW-NB15	Accuracy, Precision, Area Under Curve (AUC).
Ren <i>et al.</i> [61]	CNN	Traffic Analyses	KPI, Yahoo, Microsoft Production Data	Accuracy, Efficiency, Generality.
Wang <i>et al.</i> [62]	CNN	Network Security	USTC-TFC2016	Detection Accuracy, Precision, Recall, F1 score.
Fotiadou <i>et al.</i> [63]	CNN, RNN	IoT	KDD Cup 1999	Detection Rate, False Alarm Rate.
Torres <i>et al.</i> [64]	RNN	IoT	KDD Cup 1999	Detection Rate, False Alarm Rate.

PCA serves as a fundamental dimensionality reduction technique in IDSs [70]. Abdulhammed *et al.* employed PCA to reduce feature dimensionality before applying classifiers, such as RF Bayesian Network, Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA). Similarly, Liu *et al.* [71] combined PCA with neural networks to develop a hierarchical IDS addressing the limitations of single-level systems that typically detect anomalies exclusively.

Auto-encoders are specialized neural networks designed to reconstruct input data from unlabeled samples. Their architecture comprises three main components: an encoder, a middle layer, and a decoder. The encoder compresses input data into a lower-dimensional representation, while the decoder reconstructs the original data from this compressed form [81]. Deep auto-encoders, featuring multiple hidden layers, can capture complex and nonlinear characteristics across various applications [77]. To manage computational requirements, these networks can be trained incrementally, layer by layer.

Boltzmann Machines are generative neural networks that learn probability distributions of unseen data using reference datasets [82]. Their distinctive bidirectional connections enable bilateral information processing through a visible layer that contains known inputs and a hidden layer that houses hidden variables [83]. Restricted Boltzmann Machines (RBMs), variants of Boltzmann Machines have found applications in feature learning, dimensionality reduction, classification, and collaborative filtering. Fiore *et al.* [84] introduced a Discriminative RBM (DRBM) for network traffic anomaly detection. DRBMs enhance the original RBM framework by incorporating discriminative learning for classification tasks.

DBNs address the challenge of weight initialization in deep neural networks through a stack of RBMs serving as multiple hidden layers. Their architecture implements layer-wise unsupervised learning for effective pre-training and supervised learning techniques for network fine-tuning. Chen *et al.* [80] demonstrated DBNs' effectiveness in malware detection within edge computing environments. They excel at capturing complex relationships and learning high-level representations from large amounts of data. DBNs can extract meaningful features by leveraging their hierarchical structure and employing unsupervised pre-training approaches, to facilitate intrusion detection in IoRT systems. The discussed unsupervised models are summarized in Table 4.

5.3. Hybrid and ensemble learning algorithms

Hybrid and ensemble learning techniques enhance the effectiveness and resilience of conventional machine learning methods by combining multiple approaches. These methods have demonstrated significant improvements in system performance and a reduction in false alarm rates. Hybrid models integrate multiple learning approaches with optimization and soft computing methods, such as genetic algorithms and fuzzy logic [85]. A common hybrid

Table 4. Unsupervised machine learning-based IDSs.

References	Algorithms	Domain	Datasets	Evaluation Metrics
Peng <i>et al.</i> [72]	K-Means, PCA	Cloud computing, Social Networks	KDD Cup 1999	Compactness-Heterogeneity (CH) index, Clustering Time.
Zhang <i>et al.</i> [73]	K-Means, Deep Autoencoder	WSNs	KDD Cup 1999	Detection Rate, FPR.
Zhou <i>et al.</i> [74]	Deep Autoencoder	IoT	MNIST	Precision, Recall, F1 Score.
Thakur <i>et al.</i> [75]	Deep Autoencoder	CPSs	CICIDS2017	Precision, Recall, Specificity, F1 score.
Shone <i>et al.</i> [76]	Nonsymmetric Deep Autoencoder	Computer Networks	KDD Cup 1999	Accuracy, Precision, Recall, False Alarm and F1 Score.
Alom <i>et al.</i> [77]	DBN	Computer Network	NSL-KDD	Accuracy, Training Time
Aldwairi <i>et al.</i> [78]	RBM	Computer Networks	ISCX	Accuracy.
Elsaedy <i>et al.</i> [79]	RBM	Smart Cities	Smart water Distribution Plant Data	Accuracy, F1 Score.
Chen <i>et al.</i> [80]	DBN	Mobile Edge Computing(MEC)	MEC Environment Data	Accuracy, True Positives, True Negatives.

architecture involves using unsupervised learning for data preprocessing and feeding it into supervised learning models for classification. For example, Shahin *et al.* [86] developed a sophisticated hybrid model for attack detection in industrial IoT devices. Their approach combines an Attention-based Long Short-Term Memory (ALSTM), a Fully Convolutional Neural Network (FCN), and Gradient Boosting to accomplish the task.

Ensemble learning leverages multiple models' collective intelligence through three primary techniques: bagging, boosting, and stacking [85,87]. These approaches enhance accuracy, robustness, and generalization capabilities by exploiting model diversity [88].

Bagging (Bootstrap Aggregating) trains multiple instances of the same learning algorithm on different training datasets, created through bootstrap sampling [89]. This approach reduces overfitting and variance in the final model. Boosting sequentially improves weak learners' performance by assigning higher weights to misclassified instances, having each model correct its predecessor's error, and computing final predictions as weighted sums of the individual model outputs [90]. AdaBoost is an example of a boosting algorithm. Stacking combines multiple models' predictions by training a meta-model instead of using simple averaging or voting [91]. This approach often incorporates diverse base models such as DT, SVMs, and Neural Networks.

Several researchers have successfully implemented these techniques in IoT security. Illy *et al.* [96] developed an ensemble learner by combining DT, KNN, and Multi-Layer Perceptron classifier (MLP) to increase detection accuracy in a fog-to-things environment. Their proposed architecture included two levels of classification. The first level focused on anomaly detection to reduce classification latency, while the second level performed attack classifications. Hariharan *et al.* [97] also

proposed a stacked ensemble learning model for intrusion detection in wireless networks. Their generic framework combines RF and gradient boost as the base learners to enhance the effectiveness of their IDS. Verma *et al.* [98] proposed an Ensemble Learning-based Network Intrusion Detection System (ELNIDS) for detecting routing attacks in IoT. Their approach employed Boosted Trees, Bagged Trees, Subspace Discriminant, and RUSBoosted Trees classifiers to detect routing attacks. The results showed that the ensemble of Boosted Trees achieved the highest accuracy of 94.5%, while the Subspace Discriminant method had the lowest accuracy of 77.8%. The ensemble of RUSBoosted Trees achieved the highest value of 0.98 for the area under the ROC curve, whereas the ensemble of Subspace Discriminant had the lowest value of 0.87. These examples demonstrate the effectiveness of ensemble learning techniques in improving predictive performance in intrusion. These algorithms are summarized in Table 5.

5.4. Semi-supervised learning algorithms

Semi-supervised learning algorithms bridge the gap between supervised and unsupervised learning techniques. By leveraging labeled and unlabeled data during training, these algorithms aim to overcome the limitations of traditional learning methods [109].

Traditional machine learning approaches face distinct challenges. Supervised learning algorithms offer high accuracy but are computationally expensive and require extensive labeled data. In contrast, unsupervised learning algorithms are less precise but do not depend on costly labeled data.

Several studies have utilized semi-supervised learning for IDSs across various domains [102,103,110–112]. Ashfaq *et al.* [110] introduced an IDS based on fuzziness for semi-supervised learning. They used a Single Hidden Layer Feed-forward Neural Network (SLFN) to output a fuzzy membership vector. The SLFN assigns each sample to different categories based on their degree of membership (low, mid, or high fuzziness), and unlabeled samples are categorized into these fuzzy categories using the quantity obtained from the SLFN.

Semi-supervised learning has found applications in Intrusion Detection Systems (IDSs) in the IIoT domain as well [46,106,107]. Aouedi *et al.* [46] developed a federated semi-supervised learning approach for IIoT that harnesses both labeled and unlabeled data.

Transformers have emerged as a promising semi-supervised learning paradigm, characterized by self-attention mechanisms, elimination of recurrent units, and reduced training time compared to traditional RNNs.

Ferrag *et al.* [108] proposed SecurityBERT, a novel architecture that leverages the BERT model for intrusion detection in IoT networks. They leverage Bidirectional Encoder Representations from Transformers (BERT) and incorporate a novel Privacy-Preserving Fixed-Length Encoding (PPFLE) method. Their

Table 5. Hybrid and ensemble machine learning-based IDSs.

References	Hybrid/ Ensemble	Algorithms	Domain	Datasets	Evaluation Metrics
Alhaidariet <i>al.</i> [92]	Hybrid	K-Means, Hidden Markov Model	IoT	IoT POT	Accuracy, Error Rate, Recall, AUC.
Hong <i>et al.</i> [93]	Hybrid	SVM, Hierarchical Clustering	Internet technologies	NSL-KDD, KDD Cup 1999	Accuracy.
Li <i>et al.</i> [94]	Hybrid	RNN, RBM	Traffic Detection	ISCX-2012, DARPAR1998	Accuracy, Recall, FPR.
Benaddi <i>et al.</i> [55]	Hybrid	PCA, fuzzy Clustering, KNN	WSNs	NSL-KDD, KDD Cup 1999	Accuracy, FPR.
Ghosh <i>et al.</i> [49]	Hybrid	Fuzzy C-Means, KNN	Cloud Computing	NSL-KDD	Accuracy.
Shahin <i>et al.</i> [86]	Hybrid	LSTM, CNN	IIoT	TON_IoT	Accuracy, Precision, Recall, F1 score.
Gebremariam <i>et al.</i> [95]	Hybrid	RF, K-Means, Extreme Gradient Boost	WSNs	NSL-KDD, UNSW_NB15, CICIDS2017	Accuracy, Precision, Recall, F1 score.
Illy <i>et al.</i> [96]	Ensemble	DT, KNN, MLP	Fog-to-Things Environment	NSL-KDD	Accuracy.
Rajadurai <i>et al.</i> [97]	Ensemble	RF, Gradient Boost	WSNs	NSL-KDD	Accuracy, Precision, Detection Rate, Recall.
Verma <i>et al.</i> [98]	Ensemble	Boosted Trees, Bagged Trees, Subspace Discriminant, RUSBoosted Trees	IoT	RPL-NIDDS17	Accuracy, ROC.
Otoum <i>et al.</i> [99]	Ensemble	RF, Density Based Spatial Clustering, RBM	WSNs	NSL-KDD	Accuracy, Detection Rate, FPR.
Rajagopal <i>et al.</i> [91]	Ensemble	Logistic Regression, KNN, RF, and SVM	WSNs	UNSW NB-15, UGR'16	Accuracy, Precision, Recall, F1 Score, False Alarm Rate.
Rohini <i>et al.</i> [100]	Ensemble	RF, kNN, SVM, LSTM, GRU	IoT	RPL-NIDDS17	Accuracy, Error Rate, Recall, TNR, FPR, F1 score, MCC.
Almotairi <i>et al.</i> [101]	Ensemble	NB, RF, kNN, SVM	IoT	TON_IoT	Accuracy, Precision, Recall, F1 score.

experimental results show SecurityBERT to achieve a 98.2% accuracy in detecting 14 distinct attack types. It outperforms previous CNN-LSTM models, with an inference time of less than 0.15 seconds. These discussed algorithms are summarized in Table 6.

5.5. Reinforcement learning algorithms

Reinforcement learning (RL) algorithms learn from interacting with their environment feedback. They are useful in cases where complete knowledge or labeled data is unavailable [113]. Unlike traditional supervised or

Table 6. Semi-Supervised Machine Learning-Based IDSs.

References	Algorithms	Domain	Dataset	Evaluation Metrics
Gao <i>et al.</i> [102]	Fuzziness	Robotic System	NSL-KDD	Accuracy
Li <i>et al.</i> [103]	Fuzziness	IoT	NSL-KDD	Accuracy.
Duong <i>et al.</i> [104]	K-Means, PCA	Network Traffic Analyses	NSL-KDD	Detection Rate, FPR, Precision, and Recall.
Wagh <i>et al.</i> [105]	Self Learning Algorithm	Network Traffic Analyses	KDD Cup 1999	Accuracy, Detection Rate.
Aouedi <i>et al.</i> [46]	Semi-Supervised FL	IIoT	SCADA Dataset	Accuracy, F1 score.
Naeem <i>et al.</i> [106]	Semi-Supervised FL	IIoT	CSE-CIC-IDS2018	Accuracy
Mendonça <i>et al.</i> [107]	Sparse Evolutionary training	IIoT	DS20S, CICIDS2017	Accuracy, Precision, Recall, F1 score.
Ferrag <i>et al.</i> [108]	BERT	IoT	Edge-IIoTset	Accuracy, Precision, Recall, F1-Score, Converging Time, CPU Memory.

unsupervised learning methods, these algorithms create adaptive, self-governing agents capable of making optimal decisions that excel in scenarios where complete knowledge or labeled data is unavailable. Their adaptive nature is suited for real-time adversarial environments, such as cybersecurity applications. They have demonstrated effectiveness in various security domains, including phishing detection, host-based intrusion detection, and DDoS attack mitigation [114].

The core principle of reinforcement learning involves an agent that interacts with an environment, receives feedback in rewards or penalties, and learns to optimize its decision-making strategy through iterative interactions.

Benaddi *et al.* [55] proposed a combination of distributional reinforcement learning (DRL) and Generative Adversarial Networks (GANs) to enhance the effectiveness and resilience of IDS in IIoT environments. Tharewal *et al.* [115] also proposed a DRL-IDS system that incorporates a feature selection approach using the Light Gradient Boosting Machine (LightGBM). They used Proximal Policy Optimization (PPO2) and a multilayer perceptron as the shared network structure for both the value and strategic network. Additionally, they used the Rectified Linear Unit (ReLU) activation function and PPO2 algorithm to create the intrusion detection model.

Reinforcement learning provides a dynamic and adaptive approach to addressing complex security challenges in IIoT environments. By enabling systems to learn and adapt in real-time, RL algorithms offer a powerful mechanism for detecting and responding to emerging security threats. The key strengths of RL in IIoT security include:

- Ability to handle uncertain and dynamic environments
- Capacity for continuous learning and adaptation
- Potential to develop proactive security strategies
- Effectiveness in scenarios with limited prior knowledge

The discussed RL algorithms are summarized in Table 7.

Table 7. Reinforcement learning-based IDSs.

References	Algorithms	Domain	Dataset	Evaluation Metrics
Ferdowsi <i>et al.</i> [116]	LSTM	Autonomous Vehicle Systems	Simulation Data	Spacing Deviation.
Behzadan <i>et al.</i> [117]	Adversarial DRL	Autonomous Vehicle Systems	Simulation Data	Convergence Rate.
Benaddi <i>et al.</i> [55]	DRL, GAN	IIoT	DS20S	Accuracy, Precision, Recall, F1 score.
Tharewal <i>et al.</i> [115]	LightGBM, PPO2, MLP	IIoT	IIoT Dataset	Accuracy, Precision, Recall rate, F1 score.
Quyen <i>et al.</i> [118]	RL, GAN	IIoT	Kitsune Network Attack Datasets	Accuracy.
Gülmez <i>et al.</i> [119]	DRL	Smart Autonomous Systems	NSL-KDD, UNSW-NB15	Accuracy, Precision, Recall, F1 score.
Sethi <i>et al.</i> [120]	DRL	Cloud Infrastructures	UNSW-NB15	Accuracy, FPR.
Lopez <i>et al.</i> [121]	DRL	Modern Data Networks	AWID, NSL-KDD	Accuracy, F1 Score, Precision, Recall.
Caminero <i>et al.</i> [122]	AE-RL	Data Networks	NSL-KDD	Accuracy, F1 Score, Precision, Recall.
Otoun <i>et al.</i> [123]	RL	WSNs	KDD Cup 1999	Detection Rate, Accuracy, AUC, False Negative Rate, F1 score.

5.6. Generative adversarial networks (GANs)

Generative Adversarial Networks (GANs) address the generative modeling problem by learning the underlying probability distribution of the training data samples [124]. Initially proposed for unsupervised learning, GANs have expanded their applicability to reinforcement learning and semi-supervised learning problems [125].

The core architecture of GANs consists two key components: a generator and a discriminator. The generator creates synthetic data instances that aim to mimic real-world samples, while the discriminator evaluates and distinguishes between authentic and generated data. The adversarial approach makes GANs valuable for tasks such as data augmentation and intrusion detection.

Several researchers have leveraged GANs in cybersecurity applications. For instance, Hiromoto *et al.* integrated GANs with Distributed System Coordination and Cryptographic Hardware Monitoring for malware detection in the supply chain. Similarly, Kim *et al.* [126] proposed a transferred deep-convolutional GAN (tDCGANs) for malware detection.

5.7. Non-domain specific machine learning-based IDSs

Non-domain-specific machine learning-based IDSs are also significant to our study. Unlike their domain-specific counterparts, they are designed to identify and mitigate a broad spectrum of cyber threats without being constrained to a specific domain [111,127]. This section explores some of these models.

DT-based approaches have shown promising results. Moon *et al.* [52] proposed a behavioral analysis DT-based IDS for detecting APT, achieving an 84.7% detection accuracy.

Kabir *et al.* proposed a wrapper approach using a Bayesian network for intrusion detection. Their framework achieved a 98.2653% accuracy, with a minimal error rate of 1.73 and FPR of 0.007 on the NSL-KDD dataset. Complementing this, Farrahi *et al.* [133] developed a hybrid method combining k-means clustering with Naïve Bayes, SVM, and OneR algorithms attaining a 99.50% detection rate.

SVM approaches have also demonstrated significant potential. Heba *et al.* [48] implemented an SVM-based IDS with PCA for feature selection, focusing on reducing computational overhead while maintaining high performance, with an F1-score of 0.97.

The field has seen further advancement with semi-supervised frameworks. Yao *et al.* [111] proposed the Multilevel Semi-Supervised ML (MSML) framework, which includes modules for cluster extraction, pattern discovery, fine-grained classification, and model updating. Their approach demonstrated superior performance in accuracy, F1 score, and unknown pattern recognition. These algorithms are summarized in Table 8.

5.8. Federated learning for security and privacy

Federated learning emerged as a promising approach in intrusion detection to address data privacy and protection issues. This distributed learning paradigm enables model training across decentralized devices while keeping sensitive data localized.

In IoT systems, Li *et al.* [45] proposed the FedLog scheme for anomaly detection, implementing a Temporal Convolutional Network-Attention Mechanism-based model for system logs feature extraction. Their approach incorporates a masking strategy based on the lottery ticket hypothesis to enable communication-efficient federated Learning and to handle heterogeneous data sets. Similarly, Aouedi *et al.* [46] developed a Federated Semi-Supervised Learning (FSSL) scheme for IIoT intrusion detection. Their model uses AutoEncoders on each device for feature extraction. Their model achieves high detection accuracy while maintaining data privacy and minimal low communication overhead.

Wang *et al.* [141] proposed a two-phase hierarchical federated learning approach for anomaly detection in the IIoT as well. Their approach uses a deep reinforcement learning (DRL) algorithm to ensure data privacy and high throughput with low latency. They incorporated the Deep Deterministic Policy Gradient (DDPG) algorithm for the local model. Rashid *et al.* [142] advanced IIoT network security through a federated learning framework combining CNN for feature extraction and LSTM for classification. Additionally,

Tabassum *et al.* [143] developed FEDGAN-IDS for smart IoT systems, implementing a distributed GAN network across IoT devices for classification using augmented local data.

A significant challenge in federated learning-based IDSs is the Non-Independent-Identically Distributed (non-i.i.d) nature of real-world data. Quyen *et al.* [144] addressed this through a combined GAN and reinforcement learning approach, where GAN-based data augmentation handles class imbalance while reinforcement learning optimizes client selection. Kundu *et al.* [145] also proposed the FedPlus algorithm to enhance robustness against outliers and prevent training failure through central server aggregation. Chandran *et al.* [146] introduced a Divide-and-Conquer variant of the popular FedAvg with a Cosine Distance-based Weight metric, dividing global deep neural networks into agnostic initial layers and class-specific deep layers.

While federated learning effectively enhances privacy and reduces communication overheads, the common assumption of a trusted central aggregation node presents potential security vulnerabilities. Recent research has focused on incorporating advanced security measures, particularly homomorphic encryption, and differential privacy to protect data privacy during aggregation [147–149].

Several researchers have proposed enhanced secure federated learning frameworks. Abdel-Basset *et al.* [147] developed the Federated Graph Convolutional Recurrent Network Fed-GCRN framework for traffic flow forecasting in the Internet of Vehicular Things environment (IoVT). It employs an adaptive differential privacy mechanism for better privacy-performance trade-off, and random subsampling to improve communication efficiency and scalability in traffic forecasting. Their framework uses the FedAVG algorithm to reduce uplink communication overheads. Liu *et al.* [148] also proposed D2MIF, integrating isolation forest algorithm, deep reinforcement learning, and dynamic threshold adjustment to secure federated learning in AIoT.

Li *et al.* [150] proposed the Federated Deep Learning scheme (DeepFed) for industrial applications. DeepFed uses the novel CNN-GRU-based model and a secure communication protocol based on the Paillier cryptosystem to ensure privacy due to training. Duy *et al.* [151] introduced an IDS architecture for the IoT and IIoT combining LSTM and VGG models with homomorphic encryption and differential privacy. They extended this work by proposing a differential privacy-enabled federated learning architecture (SDN-Fed-IDS) [152]. These algorithms are summarized in Table 9.

5.9. Traditional machine learning vs advanced deep learning approaches in IoRT security

In addressing the security challenges of IoRT systems, traditional machine learning and advanced deep learning approaches offer distinct advantages and limitations. Methods such as SVM, KNN, and DT provide good baseline

Table 8. Non-domain specific machine learning-based IDSs.

References	Learning Type	Algorithms	Attacks	Dataset	Evaluation Metrics
Ingre <i>et al.</i> [50]	Supervised	DT	Dos, R2L, U2R, Probe.	NSL-KDD	Detection Rate, Accuracy, False Positives Rate
Moon <i>et al.</i> [52]	Supervised	DT	APT Attacks	API Data.	Accuracy.
Kabir <i>et al.</i> [7]	Supervised	DT	DoS, R2L, U2R, Probe	NSL-KDD.	Accuracy, Error Rate, FPR.
Nazir <i>et al.</i> [128]	Supervised	RF	Intrusion Detection	UNSW-NB15	Accuracy.
Zhang <i>et al.</i> [129]	Supervised	RF	DoS, R2L, U2R, and probe	KDD Cup 1999	Detection rate, FPR.
Senthilnayagi <i>et al.</i> [56]	Supervised	KNN	DoS, R2L, U2R, and probe	KDD Cup 1999	Accuracy, False alarm Rate.
Novikov <i>et al.</i> [130]	Supervised	RBF, MLP	DoS, R2L, U2R, Probe	KDD Cup 1999	Accuracy, False Alarm Rate.
Casas <i>et al.</i> [127]	Unsupervised	Sub-Space and Density-based clustering	DoS/DDoS, Probing Attacks, Worm propagation, Buffer Overflow	KDD Cup 1999	Computational Time.
Chimphlee <i>et al.</i> [131]	Unsupervised	Fuzzy Rough Clustering	DoS, R2L, U2R,Probe	KDD Cup 1999	Accuracy, Detection Rate, False Alarm, Correlation.
Seo <i>et al.</i> [132]	Unsupervised	RBM	DoS, R2L, U2R, and probe	KDD Cup 1999	Accuracy, Precision, Recall, F1 Score
Farrah <i>et al.</i> [133]	Hybrid	K-means, Naïve Bayes, SVM, OneR	DoS, R2L, U2R,Probe	KDD Cup 1999	Accuracy, Detection Rate, False Alarm Rate.
Heba <i>et al.</i> [48]	Hybrid	PCA, SVM	DoS, R2L, U2R, Probe	NSL-KDD, KDD Cup 1999	Accuracy.
Abhaya <i>et al.</i> [134]	Hybrid	Fuzzy C-means, SVM	DoS, R2L, U2R, Probe	KDD Cup 1999	Detection rate, False Alarm Rate, Accuracy.
Khonde <i>et al.</i> [135]	Ensemble	RF, SVM, ANN, DT, KNN	Ransomware, DDoS	NSL-KDD	Accuracy, TPR, FPR.
Li <i>et al.</i> [136]	Semi-Supervised	Fuzzy Clustering	DoS, R2L, U2R, Probe	KDD Cup 1999	Detection Rate, False Detection Rate.
Yuan <i>et al.</i> [137]	Semi-Supervised	Tri-Adaboost	DoS, R2L, U2R, Probe	KDD Cup 1999	Detection Rate, FPR, Precision.
Pachghare <i>et al.</i> [138]	Semi-Supervised	Semi-Supervised Learning	DoS, R2L, U2R Probe	KDD Cup 1999	Detection Rate, False Detection Rate.
Hao Mao <i>et al.</i> [139]	Semi-Supervised	Semi-Supervised Learning	DoS, R2L, U2R, Probe	KDD Cup 1999	TPR, Error Rate.
Hara <i>et al.</i> [140]	Semi-Supervised	Auto-encoder	DoS, R2L, U2R, Probe	NSL-KDD	Accuracy, FPR, False Negative Rate.
Yao <i>et al.</i> [111]	Semi-Supervised	Multilevel Semi-Supervised Learning	DoS, R2L, U2R, Probe	KDD Cup 1999	Accuracy, F1-score.
Ashfaq <i>et al.</i> [110]	Semi-Supervised	Fuzziness	Masquerade Attack	NSL-KDD	Accuracy.

security through lower resource requirements and faster training times. These approaches are trained on known attack patterns are effective for detecting known attacks, and are well-suited for resource-constrained IoT devices. However, they often struggle with zero-day attacks and require regular manual updates to maintain effectiveness [159]. In contrast, deep learning approaches such as RL offer superior capabilities in detecting novel attacks and adapting to emerging threats through automatic feature learning and continuous model updating. These methods can handle complex attack patterns and unstructured data streams in real-time, making them valuable for IoT environments. However, they typically demand higher computational resources and may require cloud or edge computing for optimal performance. This comparison highlights crucial security challenges in IoT systems, particularly the need to balance detection accuracy with resource constraints, maintain real-time performance, and ensure scalability across growing networks. Understanding these trade-offs is essential for developing effective security frameworks that combine traditional and deep learning approaches to create robust, resource-aware security solutions for IoT systems.

6. Evaluation of machine learning-based IDSs

6.1. Evaluation metrics for existing machine Learning-based IDSs

Popular metrics used to evaluate machine learning-based IDSs are accuracy, precision, recall, F1-score, true positive rate (TPR), true negative rate (TNR), false negative rates (FNR), and false positive rate (FPR). Understanding their possible outcomes is imperative when assessing IDSs.

Accuracy: The accuracy of a machine learning model is calculated as the ratio of correct predictions to the total number of predictions. It can be expressed as: $Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)}$.

Precision: Precision measures an algorithm's performance in identifying positive instances. Precision is calculated by dividing true positives by the sum of true positives and false positives: $Precision = \frac{(TP)}{(TP+FP)}$.

Recall: Also referred to as Sensitivity or TPR measures an algorithm's ability to avoid missing positive instances. It is computed by: $Recall = \frac{TP}{TP+FN}$.

F-Score (F1 Score, F Measure): The F Score calculates the harmonic mean of recall and precision values. It's useful for assessing the overall performance of a classifier, especially when dealing with imbalanced datasets. It is calculated as follows: $F1 - Score = 2 \times \frac{Precision \times Recall}{(Precision + Recall)}$.

False Negative Rate (FNR): This is the percentage of actual positive instances incorrectly labeled as negative. It's calculated by dividing the number of undetected true conditions by the total number of actual true conditions. A low False Negative Rate is crucial in sensitive situations where missed detection can have serious consequences.

Table 9. Federated learning for security and privacy.

References	Domain	Algorithm	FL Model	Dataset	Evaluation Metrics
Abdel-Basset <i>et al.</i> [147]	IoT	GCRN	Fed-GCRN	Traffic flow data	Mean Absolute Error.
Liu <i>et al.</i> [148]	IoT	Iforest, DRL	D2MIF	Mnist, Fashion Mnist	Accuracy, Recall, F1-score.
Li <i>et al.</i> [45]	IoT	CNN	FedLog	HDFS, BGL	Lottery Ticket Network, Accuracy.
Aouedi <i>et al.</i> [46]	IIoT	Autoencoder	FSSL	SCADA Dataset	Accuracy, F1 score.
Duy <i>et al.</i> [151]	IoT	LSTM, VGG	HEFL, DPFL	CICIDS2017	Accuracy, Precision, Recall, F1-score.
Wang <i>et al.</i> [141]	IIoT	DRL	FLAD	Simulated Data	System Throughput, Average Latency, Accuracy.
Li <i>et al.</i> [150]	Industrial CPSs	CNN, GRU	ParaAggregate	Gas pipeline	Accuracy, Precision, Recall, F1-score.
Tabassum <i>et al.</i> [143]	IoT	GAN	FEDGAN	NSL-KDD, KDD Cup 1999, UNSW-NB15	Accuracy, Loss, Recall, Precision, F1-score, AUC, Convergence Rate.
Kundu <i>et al.</i> [145]	Non-Domain Specific	Logistic Regression	FedPlus	MNIST-robust, MNIST- personalized	Accuracy.
Wang <i>et al.</i> [153]	Ubiquitous environments	Deep Q-learning	Favor	MNIST dataset, Fashion MNIST, CIFAR-10	Accuracy, Communication Efficiency
Chandran <i>et al.</i> [146]	Non-Domain Specific	CNN, LSTM,	FedAvg	CIFAR10	Accuracy, Communication Efficiency.
Yonetani <i>et al.</i> [154]	Non-Domain Specific	GAN	Forgiver-First Aggregation	MNIST, Fashion MNIST, CINIC- 10	Fréchet Inception Distance(FID).
Weinger <i>et al.</i> [155]	IoT	GAN	FedAvg	TON_IoT, DS205	Accuracy, Precision, Recall, F1-score, Average Round Time.
Wang <i>et al.</i> [156]	Non-Domain Specific	CNN, LSTM	FedMA	CIFAR-10	Accuracy, Best Local Epoch, Model Growth Rate.
Taheri <i>et al.</i> [157]	IIoT	GAN	FedGAN	Drebin, Genome, Contagio	Accuracy.
Yuanet <i>al.</i> [158]	Robotic Process Automation (RPA)	CNN, SVM,	FedProx	NSL-KDD, CICIDS2017	Accuracy, Precision, FAR, Recall rate, F1 index

True Negative Rate (TNR): Also referred to as specificity estimates the proportion of correctly classified negative instances out of all actual negative instances.

Receiver Operating Characteristic Curve (ROC): ROC plots the TPR against the FPR at various classification thresholds. It comprehensively evaluates the algorithm's performance across different decision thresholds. The Area Under the Curve (AUC) value summarizes the information shown in the ROC curve, comparing the TPR with the FPR. When the AUC value is greater than 0.5, it indicates better accuracy in distinguishing between normal and abnormal observations.

6.2. Evaluation metrics for imbalanced datasets

Some suitable metrics for imbalanced dataset classification are Balanced Accuracy [160], Macro F1 Score [161], Average Precision Score [162], G-Mean [163], and the Matthews Correlation Coefficient (MCC) score [164]. Balanced Accuracy accounts for the imbalance in a dataset by calculating the arithmetic mean of sensitivity and specificity. It provides a balanced view of the model's performance across different classes. The Macro F1 score calculates the F1 score for each class independently and then takes the average, giving equal weight to each class. The average precision score is useful when there is a significant class imbalance. It considers the precision-recall performance across different thresholds by calculating the area under the Precision-Recall curve. G-mean combines sensitivity and specificity into a single performance measure. It is useful for evaluation in scenarios where it is necessary to reduce false positives and negatives.

6.3. Evaluation metrics for IoRT

IoRT introduces characteristics and challenges that differ from the IoT environments. These differences necessitate tailored evaluation metrics to effectively measure the efficiency and reliability of machine learning-based IDSs for IoRT systems.

Heterogeneity Handling: New evaluation metrics must account for the device and data heterogeneity to ensure the IDS can handle the diverse characteristics and behaviors of different devices.

Convergence Rate: IoRT systems operate in real-time dynamic environments where timely and accurate intrusion detection is crucial. Metrics such as convergence speed and latency are essential to evaluate how quickly and efficiently machine learning-based IDS can respond to threats in dynamic settings.

Resource Utilization: Evaluation metrics should consider computational complexities in terms of, resource utilization for CPU, memory, and energy consumption, and their efficiency to ensure they are suitable for resource-constrained environments.

Mobility Impact: IoRT systems involve mobile robots moving across different networks leading to challenges like handover delays and network overhead. Handover delays and network overhead are important to evaluate the impact of mobility on the performance and reliability of machine learning-based IDSs.

Privacy Preservation: Machine learning-based IDS in IoRT environments may handle sensitive data. Privacy-preserving metrics such as privacy budget and communication rounds are important to ensure that IDSs protect sensitive information. Privacy is measured in terms of differential privacy.

*Robot Operations:*IoRT systems include robotic operations such as motion planning, physical manipulation, and multi-robot coordination. Evaluation metrics should assess the impact of IDSs on these operations, ensuring that they do not hinder robot performance and functionality.

7. Research challenges and future directions

There are some challenges researchers may face when implementing machine learning-based IDSs for IoRT systems. Some challenges include the unavailability of quality benchmark datasets, the trustworthy IoRT virtual machine environment, adversarial inputs, detection accuracy, and time complexity, new and evolving attacks, resource requirements, and post-quantum readiness. The challenges presented in implementing machine learning-based IDSs for IoRT systems provide several opportunities for future research. These challenges are displayed in Figure 6.

7.1. Developing high-quality benchmark datasets

There is a significant need for high-quality, labeled datasets that reflect the complex, heterogeneous nature of IoRT traffic. Future research should focus on creating comprehensive benchmark datasets representing diverse attack categories and include real-world scenarios. Researchers could also explore synthetic data generation methods that address data privacy concerns while ensuring that models are trained on diverse and high-fidelity data.



Figure 6. IoRT research Challenges.

7.2. Enhancing the security of IoRT virtual machine environment

Virtual machines (VMs) and robotic navigation systems enable autonomous decision-making in IoRT [165]. Given the vulnerabilities introduced by VMs, future work should explore advanced security mechanisms tailored to the virtualization layer. Researchers should investigate methods to isolate the VM environment from potential exploitation and attacks targeting the VM hypervisor while enabling real-time decision-making and autonomous navigation. Trusted Execution Environments (TEES) or secure enclaves may also be explored to enhance trustworthiness.

7.3. Defending against adversarial inputs

Adversarial machine learning is an evolving field, and future research should focus on developing robust defense mechanisms that can generalize across different types of attacks. They are manipulations of input data meant to deceive machine learning models into making misleading predictions [166]. Solutions such as adversarial training, where models are trained on normal and adversarial samples, or using ensemble models could provide more comprehensive security. Additionally, ongoing efforts are needed to create algorithms resilient to sophisticated adversarial inputs in dynamic IoRT environments.

7.4. Balancing detection accuracy and time complexity

Achieving optimal detection accuracy while maintaining low time complexity is critical for real-time IoRT applications. Future research should focus on developing lightweight and efficient models tailored to IoRT systems. Hybrid models that combine the strengths of multiple machine learning algorithms, such as CNNs and RNNs, could solve various security threats, balancing detection performance and computational feasibility. Novel approaches like edge computing could also be explored to reduce latency by processing data closer to the source.

7.5. New and evolving attacks

As the threat landscape evolves, future research must focus on creating adaptive machine learning models capable of detecting novel attack techniques. Techniques like continual learning or transfer learning could be explored to enable IDSs to adapt to new attacks without requiring full retraining. Researchers may also investigate automated threat intelligence systems that allow IoRT devices to share knowledge of new attacks without compromising privacy or performance.

7.6. Resource requirements

As resource constraints are a significant challenge for many IoRT systems, future research could focus on developing energy-efficient machine learning models. Techniques such as model compression, pruning, or quantization can reduce the computational overhead while maintaining detection accuracy. Federated learning and edge-based machine learning models should also be explored to minimize the need for centralized resources, making the IDS framework more scalable and accessible.

7.7. Post-quantum readiness

The advent of quantum computing presents both a challenge and an opportunity for IoRT security. Future research should investigate integrating post-quantum cryptographic algorithms into resource-constrained IoRT systems. These algorithms must be lightweight and efficient, balancing strong security guarantees with the limited computational power and energy availability in IoRT devices [167]. Researchers should also explore hybrid cryptographic schemes that provide quantum resistance while maintaining backward compatibility with current cryptographic standards.

8. Proposed security framework

As already established from our study, integrating diverse robotic devices into IoT introduces new security challenges that IDSs for IoT may not adequately address. Some IoRT security challenges are device and data heterogeneity, real-time operation requirements, resource constraints, mobility, and sensitive data management. The diversity of communication protocols and data produced complicates the development of a unified security solution. Furthermore, many IoRT applications are time-sensitive and require rapid detection and response to security threats to ensure safety and seamless operation. The mobility introduced by autonomous mobile robots also increases the complexity of designing security solutions that can handle the addition and removal of robotic devices in real time. However, existing machine learning-based IDSs may not be optimized for these real-time requirements. Moreover, many IoRT devices have limited computational power, memory, and energy resources, and may not accommodate some existing solutions. Assistive robots may collect sensitive data that must be protected from unauthorized access. Therefore it is essential to develop lightweight-privacy-preserving IDS solutions to resolve these challenges.

8.1. Framework components

We propose a lightweight federated learning-based IDS framework to address computational complexities and privacy-preserving issues in IoRT systems. This framework leverages lightweight models and federated learning with differential privacy to enhance security while preserving data privacy. It leverages CNNs, RNNs, and efficient transformers to handle diverse data types and improve detection accuracy while protecting data privacy. Our initial conceptual security framework is displayed in Figure 7.

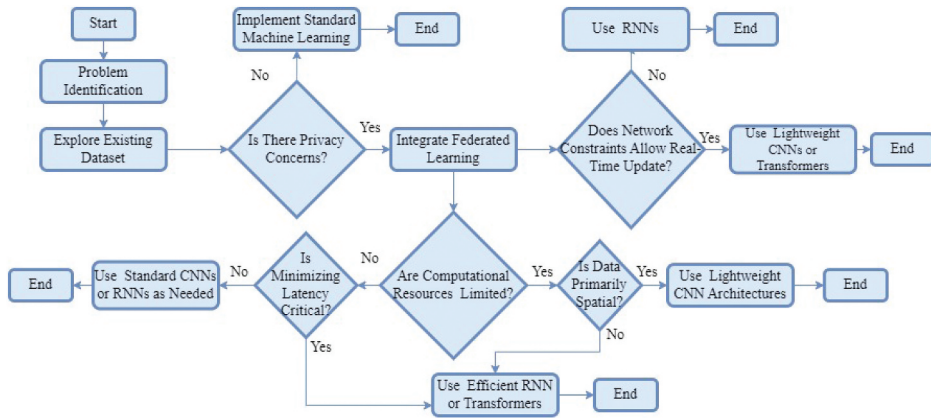


Figure 7. Initial conceptual Framework.

8.1.1. Dataset

To our knowledge, there are no existing IoRT benchmark datasets. Therefore we propose using the ROSIDS23 dataset, a network intrusion detection dataset for robot operating systems (ROS) [168]. It comprises ROS-based cyber-attacks to provide fidelity data for robotic system security research. ROSIDS23 dataset contains benign and various attack traffic collected from the ROS middleware using the tcpdump network protocol analyzer. It includes four security attacks on ROS middleware, three of which exploit the security weaknesses of ROS. They include unauthorized publishing, unauthorized subscription, subscriber flood, and DoS. These attacks target the availability of ROS middleware nodes. The dataset, collected from autonomously controlled robotic system network traffic in pcap format is suitable for intrusion detection studies in ROS-based environments.

8.1.2. Machine learning models

We propose employing CNNs, RNNs, and efficient transformers known for their abilities to extract relevant features from complex data effectively. CNNs can

capture spatial relationships between different parts of input data such as position, context, and arrangement. Cameras and LiDAR sensors provide visual data that can be effectively processed with CNNs. Additionally, they focus on learned features rather than raw data which can be beneficial in preserving privacy in IoRT systems. Lightweight CNN architectures can be optimized for resource-constrained IoRT environments.

RNNs may be employed to model sequential network data with temporal dependencies for pattern recognition and prediction in IoRT systems. They may be suitable for intrusion detection in the IoRT because IoRT systems generate sequential data from sensors, actuators, and other communication components. Furthermore, IoRT systems operate in dynamic environments, encountering varying conditions and attack types. GRUs are efficient variants that can reduce computational complexities and enhance performance in resource-constrained IoRT environments.

Unlike RNNs, transformers use an attention mechanism to simultaneously extract dependencies from input data. This parallelism helps to speed up training making them suitable for real-time intrusion detection in IoRT systems where timely responses are crucial. However, the attention mechanism requires high computational resources and makes them unsuitable for robots with onboard processing. Therefore efficient transformers that reduce running time by replacing the attention mechanism may be implemented.

8.2. Practical examples and real-world applications

Our proposed lightweight federated learning-based IDS framework addresses various challenges specific to IoRT systems, including resource constraints, real-time operational needs, and privacy concerns. Below, we provide practical examples of how this framework can enhance IoRT security in different real-world scenarios.

- (1) **Healthcare Robotics:** Assistive robots in hospitals monitor patients and aid in surgeries, often collecting sensitive health data. The lightweight CNN-based intrusion detection model within our framework can process camera and LiDAR data in real-time to detect unauthorized access or tampering with patient data. The use of CNNs ensures quick feature extraction from image data. Additionally, federated learning allows multiple hospital systems to collaborate on training the IDS without sharing raw patient data, further safeguarding privacy, though mechanisms like differential privacy must be included.
- (2) **Autonomous Drones for Delivery Services:** Delivery drones operate in real-time and are vulnerable to attacks such as GPS spoofing or unauthorized command injection. By employing RNNs in IDSs, the system can track sequential drone communication data to detect abnormal patterns or

unauthorized control attempts. This real-time detection is crucial in preventing collisions or loss of packages during delivery.

- (3) **Smart Manufacturing:** In a smart factory setting, robotic arms and sensors coordinate in real time to produce goods. These systems communicate frequently, making them susceptible to DDoS, and DoS attacks. As integrated in our framework, efficient transformers can process the communication data quickly, leveraging parallelism to ensure that the IDS detects any threats in time without affecting the operational flow.
- (4) **Agricultural Robots:** In an agricultural setting, robots perform tasks such as planting, harvesting, and monitoring crop health, often in remote locations with limited connectivity. The framework's federated learning aspect ensures that robots can locally detect security threats, such as tampered sensor readings, by training models on their data. This local detection, combined with lightweight CNN architectures, helps reduce energy consumption and computational costs. While robots do not require constant internet connectivity for local inference and threat detection, periodic communication is necessary for model updates during the federated learning process.

These practical applications demonstrate how the proposed security framework can be deployed in diverse IoRT environments, offering solutions that align with the challenges identified in this research.

9. Discussion and limitations

While this study provides valuable insights into the application of machine learning-based IDSs in IoRT environments, several limitations and shortcomings should be acknowledged:

- (1) **Theoretical Framework limitations:** Our proposed security framework, while comprehensive in its theoretical foundation, lacks empirical validation in real-world IoRT deployments. The absence of practical implementation data limits our ability to assess the framework's effectiveness under actual operational conditions, particularly in scenarios involving diverse robotic systems and varying network loads.
- (2) **Dataset Constraints:** The evaluation of existing machine learning-based IDSs' applicability to IoRT systems was primarily based on a literature review rather than experimental validation with IoRT-specific datasets. This limitation stems from the current scarcity of publicly available IoRT security datasets, which constrains our ability to make definitive conclusions about the effectiveness of proposed solutions.
- (3) **Privacy-Preserving Tradeoffs:** Although we introduced privacy-preserving concepts, our work does not thoroughly examine the performance trade-

offs between privacy preservation mechanisms and system responsiveness. The impact of implementing privacy-preserving techniques on the real-time requirements of IoRT systems requires more detailed investigations.

- (4) Scalability Analysis: The framework's scalability aspects were not addressed, particularly in scenarios involving large-scale IoRT deployments with hundreds or thousands of devices. This limitation makes assessing the framework's viability in enterprise-level implementations difficult.

10. Conclusion

CPSs, IoT, and IoRT represent interconnected paradigms that build upon each other to create more complex and capable systems. While the IoT emphasizes real-time communication and data exchange, CPSs integrate physical and computational components. IoRT is a subset of CPSs that combines IoT devices and robotic systems. As we navigate the complexities of IoRT systems, it is evident that ensuring its operation requires adaptive connectivity solutions and robust security measures. IoRT is built on the communication and network infrastructure of IoT and CPSs, and shares similarities in data sharing and communication and by extension some network attack surfaces. While machine learning models have been widely used in intrusion detection for IoT systems, their specific application in the context of IoRT is still an area that requires further research. Based on these similarities, some existing machine learning-based IDSs for IoT and other CPSs can be extended to the IoRT.

Despite these similarities, there are notable differences in real-time requirements, decentralized feedback control and physical manipulation, power consumption, mobility, and standardization. As a result, IoRT may require new and advanced machine learning-based IDSs tailored to suit their distinct requirements.

While machine learning models have been widely used in intrusion detection for IoT systems, their specific application in the context of IoRT remains unexplored. Our literature review also reveals a research gap in IoRT security as recent works mainly discuss the architecture, components, and challenges of IoRT systems. To provide a comprehensive overview of the current research landscape, we discuss the fundamental concepts of IoRT, its vulnerability to cyber-physical threats, and the challenges associated with securing these systems. It presents a comparative review distinguishing between IoT, CPSs, and IoRT to establish a basis for extending some existing machine learning-based IDS solutions and the need for more sophisticated solutions. It surveys existing machine learning-based IDSs for IoT and CPSs while exploring their potential extension to IoRT systems. Finally, it introduces a security framework to address the research challenges identified and discussed in this study.

While our proposed framework offers significant advancement in this research landscape, other areas require further investigation. Future research could explore developing adaptive learning mechanisms that dynamically adjust to evolving threats and robot behaviors in IoRT environments. Researchers may also explore robust privacy-preserving methods such as differential privacy and homomorphic encryption suitable for IoRT environments. Developing IDS solutions to handle data heterogeneity in IoRT systems is another area that could be further explored.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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